

Stage 3J Physics Note: Unit-Test Fixture Design Before C++ Implementation

GL-with-AI project

June 14, 2026

1 Scope and Status

Stage 3J is a design stage. It does not implement C++ tests, finite differences, or modified-cartoon CCZ4 source terms. Its purpose is to define the fixture layer that should exist before project-specific source terms are implemented.

The target bookkeeping remains

$$\text{GR_SPACEDIM} = 4, \quad \text{CH_SPACEDIM} = 2, \quad N_{\text{hidden}} = 2.$$

The hidden variable h_{ww} is a Cartesian-like conformal metric component, not $g_{\theta\theta}$. The spherical/cartoon reconstruction supplies the external factor x^2 , and all trace and trace-free projections are full four-spatial-dimensional operations with denominator 4.

2 Fixture Classes

The fixture suite should separate exact symbolic checks, C++ algebra unit tests, RHS-block tests, reference-code comparisons, and convergence or evolution-level tests. A green result in one class should not be read as physical correctness of the whole evolution system.

Stage 3J classifies tests as exact, roundoff-level, tolerance-based, reference-based, or convergence-based. Numeric tolerances are intentionally not chosen here. Future C++ tolerances must be justified from truncation order, resolution, floating-point roundoff, interpolation or ghost-zone treatment, and reference-data uncertainty. They must not be arbitrary thresholds chosen only because a run passes. Exact symbolic identities should remain exact in the symbolic scripts.

2.1 Existing Symbolic Gates

The current repository already contains symbolic gates for:

- Stage 3C flat cartoon geometry, GP extrinsic curvature, and hidden trace multiplicity;
- Stage 3D diagonal nontrivial $\gamma_{ww} = q(x, z)$ Christoffel and Ricci checks;
- Stage 3E nonconstant $q(x, z)$ warped-product Ricci derivative-sector regression;
- Stage 3F diagonal conformal-cartoon algebra and determinant/trace guards;
- Stage 3G off-diagonal conformal-cartoon determinant, inverse, trace, K_{xz} , and diagonal-limit checks;

- Stage 3G sheared-flat off-diagonal Ricci gate;
- Stage 3I small- x removable-limit and assembled $\hat{\Gamma}^x$ axis guard.

These symbolic gates should become fixture generators or expected-value references for the future C++ tests.

2.2 Future C++ Unit Tests

The first C++ tests should isolate small blocks:

- determinant enforcement with hidden h_{ww} ;
- full 4D trace-free projection with hidden multiplicity;
- the load-bearing $1/4$ trace denominator;
- off-diagonal h_{xz} determinant and inverse handling;
- K_{ij} reconstruction, including

$$K_{xz} = \chi^{-1} \left(A_{xz} + \frac{1}{4} h_{xz} K \right);$$

- cartoon Ricci source blocks;
- a round two-sphere positive-curvature geometry anchor,

$$R_{S^2} = \frac{2}{a^2},$$

or the equivalent Ricci/scalar-curvature statement in the project's angular extraction convention;

- regularized small- x combinations;
- assembled $\tilde{\Gamma}^x$ and $\hat{\Gamma}^x = \tilde{\Gamma}^x + 2Z^x$;
- Gamma-driver and $\hat{\Gamma}^A$ ownership boundaries.

Each off-diagonal fixture should include a diagonal-limit regression. Each axis-limit fixture should include both regular Taylor data and deliberately irregular data that exposes the expected singular behavior.

3 Pre-C++ Gates

Before Stage 3K turns this design into an implementation plan, the minimum gates are:

- all Stage 3C–3I symbolic scripts pass;
- no stale stage-label or old conformal-cartoon filename references remain;
- determinant, trace, h_{ww} , and $1/4$ conventions are documented;
- $\hat{\Gamma}^A = \tilde{\Gamma}^A + 2Z^A$ is documented as the connection-sector blueprint;
- small- x axis risks, the co-located physical singularity, and turduckening/interior regularization are documented as distinct issues;
- unresolved GRChombo-facing convention choices are carried forward.

4 Reference and Evolution Anchors

Flat data are necessary but not sufficient. The future validation suite should also include sheared-flat zero Ricci/source checks, gauge-fixed uniform GP-string stationarity outside the turduckening region, a carefully scoped uniform Schwarzschild/black-string anchor, linear Gregory–Laflamme threshold/growth comparison after convention matching, constraint-violation damping injection tests, later external/reference implementation comparisons, and resolution self-convergence.

The Schwarzschild $\times S^1$ anchor is slicing-dependent. In a GP-type slice the spatial metric is flat times S^1 , while the nontrivial black-hole geometry is carried by K_{ij} . This gives a spatial-Ricci geometry check for the GP slice, not a claim that every Schwarzschild spatial slice has $R_{ij} = 0$. A full spacetime Ricci-flatness check for Schwarzschild₄ $\times S^1$ is out of scope for the current spatial-cartoon fixture layer until a spacetime-Ricci symbolic engine exists.

The GL spectrum is a required physical-sector milestone, but it is not a closed-form elementary unit test. It is a semi-analytic or literature/numerical spectral anchor whose use requires matching radius convention, periodicity, perturbation sector, gauge, extraction variable, and slicing/conformal-variable convention. It validates the physical linearized RHS, including metric, K , A_{ij} , gauge/source coupling, and cartoon-sector terms.

Constraint-damping validation is a separate required milestone. Since the physical GL mode is expected to be constraint-satisfying in the continuum, it is necessary but not sufficient for damping signs. A non-vacuous damping test should inject controlled linearized Hamiltonian, momentum, Θ , and encoded- Z^i violations around flat or uniform-string data, with

$$\hat{\Gamma}^i = \tilde{\Gamma}^i + 2Z^i.$$

Where feasible, the expected linearized CCZ4 constraint-subsystem decay rates/signs for the chosen κ_1, κ_2 , gauge, and background should be derived or documented. Otherwise this fixture remains high risk and reference/formulation-derived.

5 Unresolved Confirmations

The following remain open before implementation:

- determinant and trace-free enforcement with hidden h_{ww} in the GRChombo-facing project variable set;
- exact $\tilde{\Gamma}^x$ sign and full $\hat{\Gamma}^A$ convention in the GRChombo-facing cartoon extension;
- Gamma-driver and $\hat{\Gamma}^A$ coupling ownership in the GRChombo-facing implementation;
- gauge choice for uniform-string stationarity;
- conventions for GL dispersion comparison;
- which external/reference comparisons are term-by-term and which are only trajectory-level validation.

6 Non-Goals

Stage 3J does not implement C++ tests or source terms, choose final production tolerances, settle convention questions, or replace the Stage 3K minimal C++ implementation plan.